

Work Breakdown Structure and Execution Steps

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The hierarchy and work-package dictionary describe the project's work. Status reflects available artifacts, not an invented schedule or completed production implementation. Sujay Hegde is the owner of each work package.

1. Hierarchical WBS

- 1.0 Domain & SSL Certificate Expiry Alert System
 - 1.1 Requirements and planning
 - 1.1.1 Define actors, scope, five FRs, and two NFRs
 - 1.1.2 Model use cases and acknowledgment flow
 - 1.1.3 Produce RTM and SRS
 - 1.1.4 Organize Jira epics, backlog, and defect evidence
 - 1.2 Secure configuration and persistence
 - 1.2.1 Authentication and role authorization
 - 1.2.2 Asset registry and dashboard CRUD
 - 1.2.3 Durable records, unique alert keys, and audit events
 - 1.3 Monitoring and notification
 - 1.3.1 Daily scheduler and TLS adapter
 - 1.3.2 WHOIS adapter, parsing, timeouts, and retries
 - 1.3.3 Threshold alert lifecycle and acknowledgment
 - 1.3.4 48-hour escalation and notification delivery
 - 1.3.5 Monthly report generation and email
 - 1.4 Verification and operations
 - 1.4.1 Functional tests, 1,000-asset benchmark, HTTPS security checks
 - 1.4.2 Fault isolation, recovery, and monthly availability monitoring
 - 1.5 Assignment documentation and evidence
 - 1.5.1 Separate coffee-kiosk architectural exercise
 - 1.5.2 GitHub and Jira evidence indexing
 - 1.5.3 Actual Copilot generation and evidence capture
 - 1.5.4 Final README, PDFs, and submission review

2. Work-package dictionary

Package	Output / completion criterion	Requirement	Depends on	Current status
WP-1.1	Approved requirements, actors, use cases, flow, RTM, SRS	All seven	None	Documents prepared
WP-1.2	Jira setup and defect evidence indexed with source pages	All seven	WP-1.1	Captured setup evidence available; sprint completion not asserted
WP-2.1	Login/session security and authorization checked on every protected route	FR-003, FR-004, NFR-001	WP-1.1	Starter role guard only; production auth planned

Package	Output / completion criterion	Requirement	Depends on	Current status
WP-2.2	Asset CRUD dashboard backed by durable registry; removed asset excluded from scans	FR-004	WP-2.1	Starter registry fixture only; dashboard/persistence planned
WP-2.3	Transactional alert/scan/audit storage; duplicate and race tests pass	FR-001 to FR-005	WP-2.2	Planned
WP-3.1	Actual TLS adapter and daily job detect expiry and validation failure	FR-001	WP-2.2, WP-2.3	Injected fixtures available; live adapter/scheduler planned
WP-3.2	WHOIS parser and bounded retries handle unavailable registry data	FR-002, NFR-002	WP-2.2	Injected fixtures available; live adapter planned
WP-3.3	Alert thresholds and acknowledgment state transitions tested	FR-001, FR-002, FR-003	WP-3.1, WP-3.2	Starter logic and fixtures available; durable integration planned
WP-3.4	Escalation at 48h; actual mail delivery and retry evidence	FR-003	WP-3.3	Starter deadline logic available; mail integration planned
WP-3.5	All-asset summary; scheduled monthly delivery, identity, and retry checks	FR-005	WP-3.1, WP-3.2, WP-3.4	Starter report fixture only; scheduling/mail planned
WP-4.1	TC-001 to TC-012 executed; benchmark below 180s and HTTPS verified	Five FRs, NFR-001	WP-2.1 to WP-3.5	Core unit tests available; integration/security/performance planned
WP-4.2	Failure isolation, recovery, and measured 99.9% monthly availability	NFR-002	WP-3.2, WP-4.1	Starter failure fixture available; operational measurement planned
WP-5.1	Coffee-kiosk diagram and one-page justification exported	Separate Lab 3 handout	None	Available in folder 2
WP-5.2	Real repository/history and Jira screenshots indexed	Assignment evidence	WP-1.2	Available in folder 3

Package	Output / completion criterion	Requirement	Depends on	Current status
WP-5.3	Run GitHub Copilot on project code; retain prompt/output and real screenshot or repository link	Assignment item 5	WP-1.1	Pending actual Copilot session; Codex starter supplied
WP-5.4	All deliverables linked; PDFs readable; no missing or falsely attributed evidence	Assignment checklist	WP-5.1 to WP-5.3	Documentation prepared; Copilot evidence outstanding

3. Execution steps and review gates

1. Review the seven baseline requirements and actor permissions. Resolve production decisions in SRS section 9 before selecting adapters.
2. Implement authentication, registry CRUD, and durable storage. Gate: unauthorized mutation rejected and removed domains excluded from the next scan.
3. Integrate TLS and WHOIS adapters and the daily scheduler. Gate: controlled real endpoints produce correct expiry/error records with bounded attempts.
4. Integrate alert notifications, acknowledgment, and 48-hour escalation. Gate: thresholds are correct; duplicate daily notifications and acknowledgment/escalation races are prevented.
5. Integrate monthly report scheduling and mail. Gate: every current asset is represented, including failed checks, and report retries do not duplicate logical reports.
6. Execute the RTM verification plan. Gate: functional integration evidence, the defined performance benchmark, and HTTPS/role checks are recorded separately from prototype unit tests.
7. Deploy health monitoring and observe availability over a complete month. Gate: measured availability meets NFR-002; failures and recovery procedures are documented.
8. Complete the actual Copilot session, capture evidence, verify all links and exported PDFs, and review the assignment checklist before submission.

4. Risk and mitigation

Risk	Effect	Planned mitigation / verification
WHOIS expiry absent or rate-limited	False healthy status or missed registration warning	Preserve unknown/error state; bounded retries; TC-004, TC-013
Invalid or expired TLS certificate	Handshake failure hides expiry	Explicit validation failure; controlled endpoint integration in TC-002
Duplicate job execution	Repeated warnings or escalations	Stable keys and durable atomic state; repeat/concurrent-worker tests
Acknowledgment races escalation	Unnecessary security notification	Transactional conditional update; acknowledgment and deadline integration tests
Provider latency exceeds benchmark	NFR-001 fails	Record latency profile; bounded concurrency and timeout budget; TC-011

Risk	Effect	Planned mitigation / verification
Screenshot or generated-code attribution unsupported	Incomplete evidence submission	Preserve provenance; capture actual pages; retain actual Copilot output before marking WP-5.3 complete

No calendar deadlines or production completion percentages are asserted. The existing Scrum report's 26 planned points describe that captured sprint; they are not an estimate for this complete WBS.